

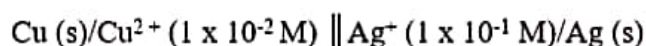
REVIEW QUESTIONS

- 1) Define the term single electrode potential. Describe the origin of single electrode potential.
- 2) What is single electrode potential? Derive Nernst equation for single electrode potential
- 3) What are ion selective electrodes? Explain determination of pH of a solution using a glass electrode.
- 4) Explain the terms with examples (i) galvanic cell (ii) half cell.
- 5) What are reactions that occur at the electrodes of a galvanic cell? Explain by taking Daniel cell as an example.
- 6) What are galvanic cells, electrolytic cells, reversible and irreversible cells?
- 7) How potential of an electrode measured using a calomel electrode?
- 8) Why the anode of a galvanic cell is negative and cathode is positive? Write different electrode reactions occur at the electrodes.
- 9) Explain the method of determination of pH of a given solution using glass.
- 10) Describe the construction of calomel electrode. Mention the advantages of this reference electrode.
- 11) What are ion selective electrodes. Give examples?
- 12) Describe the role of salt bridge in the Denial cell.
- 13) Discuss the construction of a calomel electrode with relevant chemical reactions. What are the advantages?
- 14) What are concentration cells? Derive an expression for the $e.m.f.$ of a concentration
- 15) Justify the following statements
 - (i) In a galvanic cell anode is $-ve$ and cathode is $+ve$.
 - (ii) The $e.m.f.$ of a cell is always positive.
 - (iii) A salt bridge is used in the construction of a Galvanic cell
 - (iv) Ordinary volt meter are not used in the voltage measurement of Galvanic cell.
 - (v) In a concentration cell, no electricity flow when the concentration of metal ion is same in both the half cell.
- 16) A cell is constructed by coupling Zn-electrode dipped in 0.5 M $ZnSO_4$ and Ni-electrode dipped in 0.05 M $NiSO_4$. Write the cell representation and cell reactions. Calculate $e.m.f.$ of the cell. Given: standard reduction potentials of Zn and Ni are -0.76 and -0.25 V respectively.
- 17). Calculate the voltage generated in the following cell $Fe/Fe^{2+} || Mn^{2+}/Mn$. When iron rod is immersed in 1.0 M $FeSO_4$ solution and Mn rod is immersed in 0.1M $MnSO_4$ solution. Given E^0 for Fe^{2+}/Fe is -0.4V and Mn^{2+}/Mn is -1.18V.
- 18) What is *single electrode potential*? Calculate the $e.m.f.$ of the Cell $Fe/Fe^{2+}(0.01) || Ag^+(0.1)/Ag$ at 298 K, if standard reduction electrode potentials of Fe and Ag electrodes are -0.44 and 0.8 V respectively.
- 19). A cell is formed by placing two copper electrodes immersed separately in cupric sulphate solutions of which one is 100 times more concentrated than the other. Write the cell reactions and calculate the potential at 298 K.
- 20) A concentration cell was constructed by immersing two silver electrodes in 0.05 M and 0.1 M $AgNO_3$ solution. Write cell representation, cell reactions and calculate the $e.m.f.$ of the concentration cell.

21) What is standard electrode potential? The spontaneous galvanic cell -

Tin/Tin ion (0.024 M) || Tin ion (0.64 M)/Tin develops an *emf* of 0.0126 V at 25 °C. Calculate the valency of tin.

22) Write the electrode reactions and calculate the *emf* of the following cell at 298K Given $E^\circ_{\text{cell}} = 1.3 \text{ V}$.



23) What are concentration cells? Calculate the *emf* of the following cell at 298K $\text{Ag(s)}/\text{Ag}^+ (0.01\text{M}) \parallel \text{Ag}^+ (0.1\text{M})/\text{Ag(s)}$

24). A cell uses Zn^{2+}/Zn and Ag^+/Ag electrodes. Write the cell representation, half cell reactions and net cell

reaction. Calculate the *emf* of the cell. Given $E^\circ_{\text{Zn}^{2+}/\text{Zn}} = -0.76 \text{ V}$; $E^\circ_{\text{Ag}^+/\text{Ag}} = 0.8 \text{ V}$.

25). Calculate the *emf* of a cadmium-copper cell in which Cd is in contact with 0.002 M CdSO_4 and Cu in contact with 0.02M CuSO_4 solution. The $E^\circ_{\text{cell}} = 0.74 \text{ V}$ at 298K.

Question Bank for battery Technology

1. What are batteries? Explain the classification of batteries with examples.
2. What are the principal components of a battery? Explain their working.
3. Explain the operation mechanism of a secondary battery.
4. Explain battery characteristics.
4. Explain the construction, reactions, and applications of Ni-MH battery
5. Explain the construction, reactions, and applications of Li-ion batteries.

Question Bank on Corrosion

1. Explain the mechanism of wet corrosion of iron by electrochemical theory with relevant reactions.
2. Define metallic corrosion? Discuss the following factors which influence the rate of corrosion. (i) Nature of the metal and surface state of the metal (ii) Hydrogen over voltage (iii) Inherent tendency of the metal to form protective films
3. Define metallic corrosion? Discuss the following factors which influence the rate of corrosion. (i) pH of the medium (ii) Relative areas of anode and cathode,
4. Discuss differential aeration corrosion with a suitable example.
5. Discuss stress corrosion by taking a suitable example.
6. What is anodizing? Explain anodizing of aluminum.
7. Write a note on caustic embrittlement.
8. Explain differential metal corrosion with suitable example.

9. Explain: (i) Ship sailing in water corrode below water line
(ii) Copper bolt should not be used in contact with aluminium.
(iii) Aluminium articles are self protected against corrosion while iron articles are not.
(iv) Iron corrodes faster when in contact with copper but corrodes slowly when in contact with tin
(v) Zinc in contact with silver undergoes corrosion faster than Zinc in contact with copper.
(vi) Though aluminium is with lower electrode potential than iron it has higher corrosion resistance than iron in oxidizing environments.
(vii) Iron bolts in copper vessel are undesirable.
(viii). Even if the zinc coating on iron is discontinuous, iron is free from corrosion.
(ix) Ocean going ships suffer differential aeration corrosion, but ships sunk under sea do not.
(x) Part of nail inside the wooden frame undergoes corrosion but the exposed area does not.
10. How is a metal structure protected by cathodic/anodic metal coating method?
11. Explain what type of corrosion occurs when (i) Screw and washer are made of different metals
(ii) Presence of NaOH in mild steel boiler under stress.
12. Discuss the following corrosion control methods (i) Galvanizing (ii) Tinning
13. Write a note on phosphating
14. What is cathodic protection? Discuss sacrificial anode method of prevention of corrosion.
15. Explain how corrosion is controlled by using anodic and cathodic inhibitors with suitable example and reactions.